

Literature Survey

1. Background Work

The rapid growth of social media platforms and online news sources has led to an unprecedented increase in the spread of misinformation across multiple domains, including healthcare, politics, finance, and public policy. Traditional misinformation detection systems primarily rely on text-based features and semantic similarity to distinguish between real and fake misleading information. However, these approaches often struggle when applied across domains because vocabulary, topics, and contextual information vary significantly between different areas. To elaborate consider a model that is to be used to provide information around share market, now if the model is to study news from different domains like finance, health care and others its feasible to train a specialist model for finance news only or healthcare news only but having multiple such models for detection is not feasible. we cannot train the model to check for misinformation in every domain that would affect the other target objectives of the model.

As a result, research have increasingly explored the direction of retrieval-based approaches and Large Language Models (LLMs) that can leverage contextual examples to improve misinformation detection performance.

Recent studies have shown that misinformation is not solely characterized by its textual content. But emotional characteristics, audience reactions, stance patterns, and supporting evidence often provide additional signals that help distinguish credible information from misleading claims. In particular, affect-aware retrieval methods have gained attention because emotional patterns such as fear, anger, uncertainty, and trust frequently appear across different misinformation domains. Simultaneously, research in social context analysis has demonstrated that comments, user reactions, and credibility indicators can provide valuable information beyond the original news content. These developments suggest that effective cross-domain misinformation detection may require combining multiple contextual signals rather than relying exclusively on semantic similarity.

2. Review of Existing Literature



Paper	Main Idea	Interesting Observation	Relevance to Proposed Work
RAEmoLLM: Retrieval-Augmented Emotion-Aware Misinformation Detection	Uses affective embeddings along with semantic similarity for retrieval.	Emotion patterns often transfer better across domains than textual content.	Forms the primary retrieval baseline for our work.
Emotion-Based Misinformation Detection Survey	Reviews the role of emotions, comments, stance detection, and propagation behaviour in misinformation detection.	Audience reactions often contain useful credibility signals. The support and acceptance / denial of the audience helps understand the confidence of the statement.	Motivates the use of social context information.
EmoLLMs: Affective Large Language Models	Develops emotion-specialized language models for affective reasoning tasks.	Emotion-aware models outperform generic language models on affective tasks.	Provides the foundation for extracting affective features.
Knowledge-Aware Fake News Detection (KAN)	Demonstrates the value of structured information in misinformation detection.	Relationships among claims and supporting information can improve reasoning.	Supports the idea of representing claim and evidence structures.
Knowledge-Informed Fake News Classification	Combines textual understanding with structured contextual information.	Additional contextual information improves robustness across datasets.	Motivates lightweight structured representations.
FANG: Social Context for Fake News Detection	Utilizes social interactions and surrounding context for fake news detection.	Social behaviour provides useful credibility indicators.	Supports the incorporation of social context features.

3. Identified Research Gap

The literature indicates that affect-aware retrieval significantly improves cross-domain misinformation detection by selecting emotionally relevant examples. However, most existing retrieval frameworks primarily focus on semantic and affective similarity while overlooking other forms of contextual information that may also transfer across domains. Although several studies have demonstrated the importance of comments, audience reactions, stance patterns, and credibility signals, these factors are typically investigated separately from retrieval-based approaches.

Furthermore, existing knowledge-based approaches often rely on complex knowledge graphs or entity-level representations that can be difficult to construct and maintain for dynamic information sources such as live news. While these methods improve classification performance, they often introduce significant implementation complexity without directly contributing to the retrieval process.

Another limitation of current retrieval-based misinformation detection systems is that they rely primarily on information available within the training dataset. Consequently, newly emerging news events or claims may not have closely related examples available for retrieval, reducing the effectiveness of affect-aware methods when dealing with unseen information. Moreover, most existing approaches depend on a single decision-making model, which limits transparency regarding how different sources of evidence contribute to the final prediction.

Therefore, there remains an opportunity to investigate a modular framework that combines affect-aware retrieval with dynamic external evidence retrieval and mathematically grounded evidence fusion. Such a framework would allow independent analysis of multiple information sources while providing a more transparent and explainable decision-making process.

4. Proposed Method

Based on the identified research gap, this work proposes a **Multi-Agent Evidence Fusion Framework** for cross-domain misinformation detection. The proposed system extends the RAEmoLLM framework by incorporating multiple specialized agents that independently analyze different aspects of a news article. Rather than relying solely on a Large Language Model to generate the final prediction, each agent produces measurable evidence that is subsequently fused using **Dempster–Shafer Evidence Theory (DST)**, providing a mathematically grounded and explainable decision process.

The first agent implements the original **RAEmoLLM** framework. Given an input news article, the agent extracts the affective representation by computing the Valence, Rationality, Emotion, and Global (VREG) scores. These affective embeddings are combined with semantic similarity to retrieve emotionally relevant examples from the misinformation dataset. The retrieved examples and similarity scores provide evidence regarding whether the emotional characteristics of the article resemble previously observed misinformation.

The second agent performs **external evidence retrieval**. The input article or its generated summary is submitted to an external retrieval service such as the **Tavily Search API**, which retrieves relevant articles from multiple online news sources. The retrieved documents are analyzed to determine whether they support, contradict, or provide insufficient evidence for the claim presented in the input article. Instead of directly using the retrieved articles for classification, the agent summarizes the evidence and produces an evidence consistency score based on the agreement between the retrieved sources and the original claim.

The third agent performs **claim consistency and source analysis** by extracting structured information from both the input article and the retrieved evidence. Information such as the primary claim, supporting entities, dates, organizations, and sources is organized into a structured evidence representation. This representation provides an interpretable summary of the factual information gathered from multiple sources while preserving the relationships between claims and supporting evidence.

After all agents complete their individual analyses, their outputs are converted into confidence values representing the degree of belief that the news article is either genuine or misleading. Instead of assigning fixed weights to these confidence scores, the proposed framework employs **Dempster–Shafer Evidence Theory** to combine evidence from multiple independent agents. DST provides a principled mechanism for modeling uncertainty and resolving conflicts among heterogeneous evidence sources. This allows the framework to combine emotional similarity, external evidence, and claim consistency while explicitly representing uncertainty in the final decision.

The fused belief values are then used to determine the final classification of the news article as **Real** or **Fake**, together with an associated confidence score. Because each agent contributes independently to the final belief assignment, the framework is able to provide an explainable justification for every prediction. Rather than relying solely on the reasoning capability of a Large Language Model, the final decision is supported by quantifiable evidence obtained from affective retrieval, external verification, and structured claim analysis.

The proposed framework will be evaluated by comparing its performance against existing retrieval strategies, including **random retrieval**, **semantic retrieval**, and **affect-aware retrieval (RAEmoLLM)** across multiple misinformation datasets. Evaluation will focus not only on classification performance but also on the framework's ability to provide transparent, evidence-based explanations for its predictions.

